NFC-Based Smart Attendance System with Bluetooth & USB Integration

Project Proposal

1. Introduction

The NFC-Based Smart Attendance System automates attendance tracking by allowing students to tap their smartphones (acting as NFC tags) on an NFC reader. The system verifies the student, marks their attendance in an Excel sheet, and, if absent, sends a notification via SMS Chef API. A small OLED/TFT display will also greet the student by name upon successful scanning.

This system provides two communication options for sending attendance data to a PC:

1. Wired (USB - Serial Communication)
2. Wireless (Bluetooth - HC-05 Module)

The wired option ensures a stable and direct connection, while the Bluetooth option allows wireless attendance logging without requiring an internet connection.

2. System Overview

The system consists of three primary components:

- Input: NFC data from students' smartphones.
- Processing: Data validation, attendance logging, and notification handling.
- Output: Attendance stored in Excel, SMS notifications, and a display greeting.

3. Functional Components

3.1 Input

- NFC-Enabled Smartphones (Students' phones act as NFC tags)
- NFC Reader Module (Reads NFC tags and extracts UID)

3.2 Processing

- Arduino Uno R3
    - Reads NFC data from the smartphone.
    - Matches UID with stored student information.
    - Sends attendance data to a PC via USB (Serial Communication) or Bluetooth (HC-05).

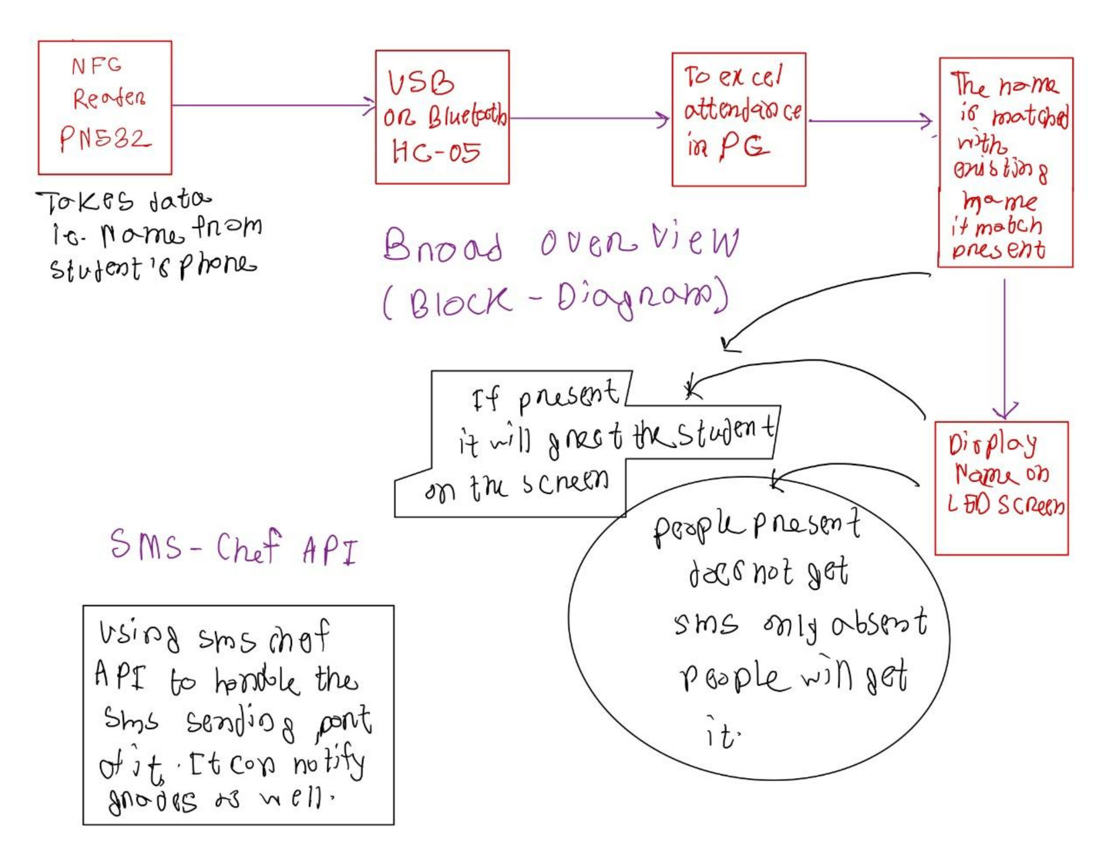
- Handles API calls for SMS notifications (if absent).

- Python Script on PC
  - Receives attendance data from Arduino via USB or Bluetooth.
  - Updates attendance records in Excel (xlsm).
  - Triggers SMS Chef API to notify students if they were absent.

3.3 Output

- OLED/TFT Display (Shows "Hello, [Student Name]")
- Excel Attendance Sheet (Logs attendance and grades)
- SMS Notification (Sent if the student is absent)

4. Block Diagram (the purple arrows indicate data traveling through Arduino to the respective components)



5. Components Required

Component	Purpose	Estimated Cost
Arduino Uno R3	Microcontroller (Handles NFC & Bluetooth)	$12.99 - $25.00
HC-05 Bluetooth Module	Sends attendance data wirelessly to PC	$5.00 - $10.00
PN532 NFC Reader Module	Reads NFC tags (phone-based attendance)	$8.99 - $39.95
OLED Display (0.96"/1.3")	Shows student name after scanning	$5.00 - $20.00
USB Cable	Programming and power	$5.00 - $10.00
Jumper Wires & Breadboard	Circuit connections	$5.00 - $10.00

Total Estimated Cost: $40 - $100 (Depending on the chosen components)

6. Conclusion

This project will create a seamless NFC-based attendance system that replaces traditional manual tracking. By integrating Bluetooth, USB, and SMS APIs, the system ensures accuracy, convenience, and flexibility for both wired and wireless attendance tracking.